



KreditKisan – AI-Powered Microloan Dynamic Pricing System

Cloud-Based FinTech Loan Risk Assessment & Interest Prediction Platform

AWS CLOUD

AI/ML

FINTECH

STREAMLIT

The Problem We're Solving

Slow & Rigid Systems

Traditional loan approvals take days with fixed, one-size-fits-all interest rates.

Unfair Pricing

Small farmers and microloan applicants receive non-personalized rates regardless of actual risk.

Manual Verification

Human-driven processes introduce delays, errors, and inconsistency in loan decisions.

No Dynamic Risk Model

Existing systems lack ML-based personalized pricing that reflects real applicant profiles.



Project Overview

KreditKisan is an end-to-end intelligent lending platform that predicts loan eligibility, calculates dynamic interest rates, and delivers instant decisions — all powered by machine learning and hosted on AWS cloud infrastructure.



Submit Application

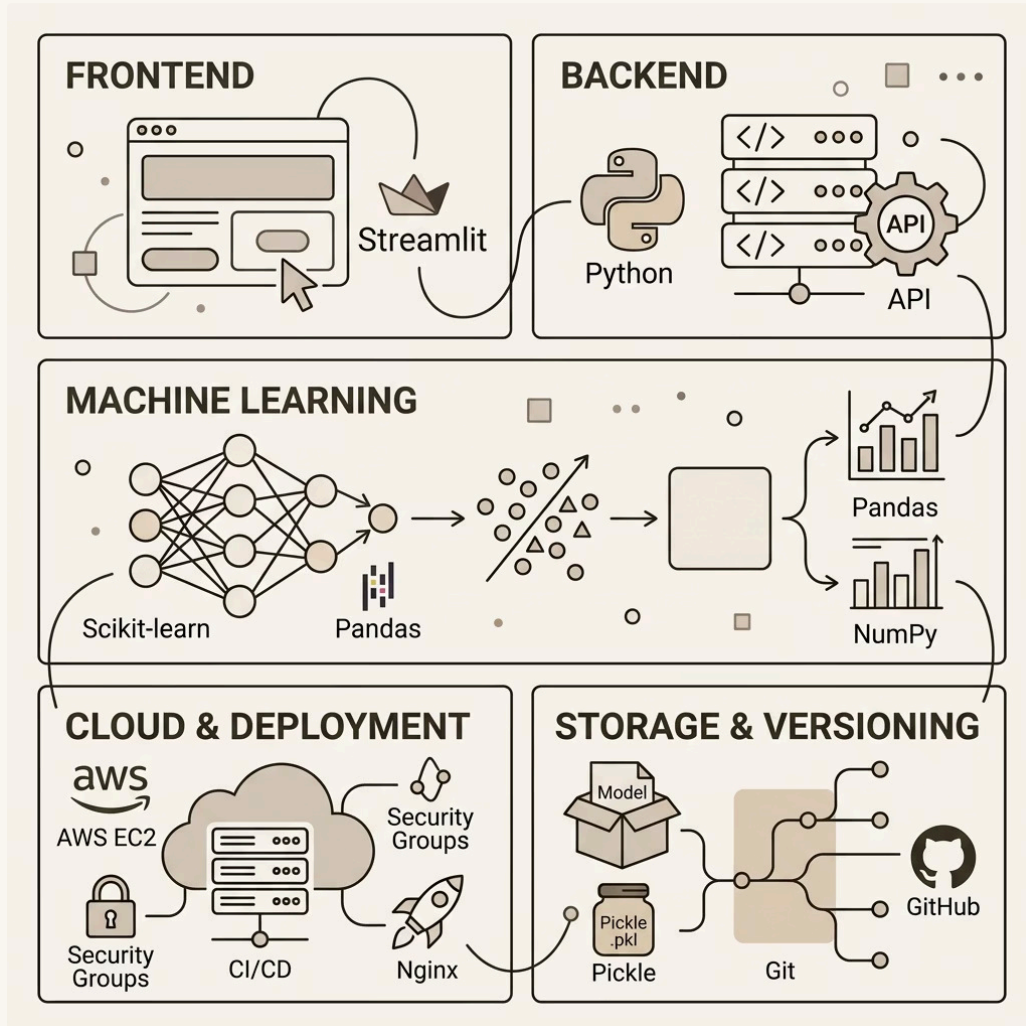
Validate Input

ML Risk Eval

Rate & Decision

The platform bridges the gap between underserved borrowers and fair, data-driven credit decisions in real time.

Technology Stack



Built for Scale & Simplicity

Every layer of KreditKisan is powered by proven, open-source technologies — from data preprocessing with Pandas to model serving on AWS EC2.

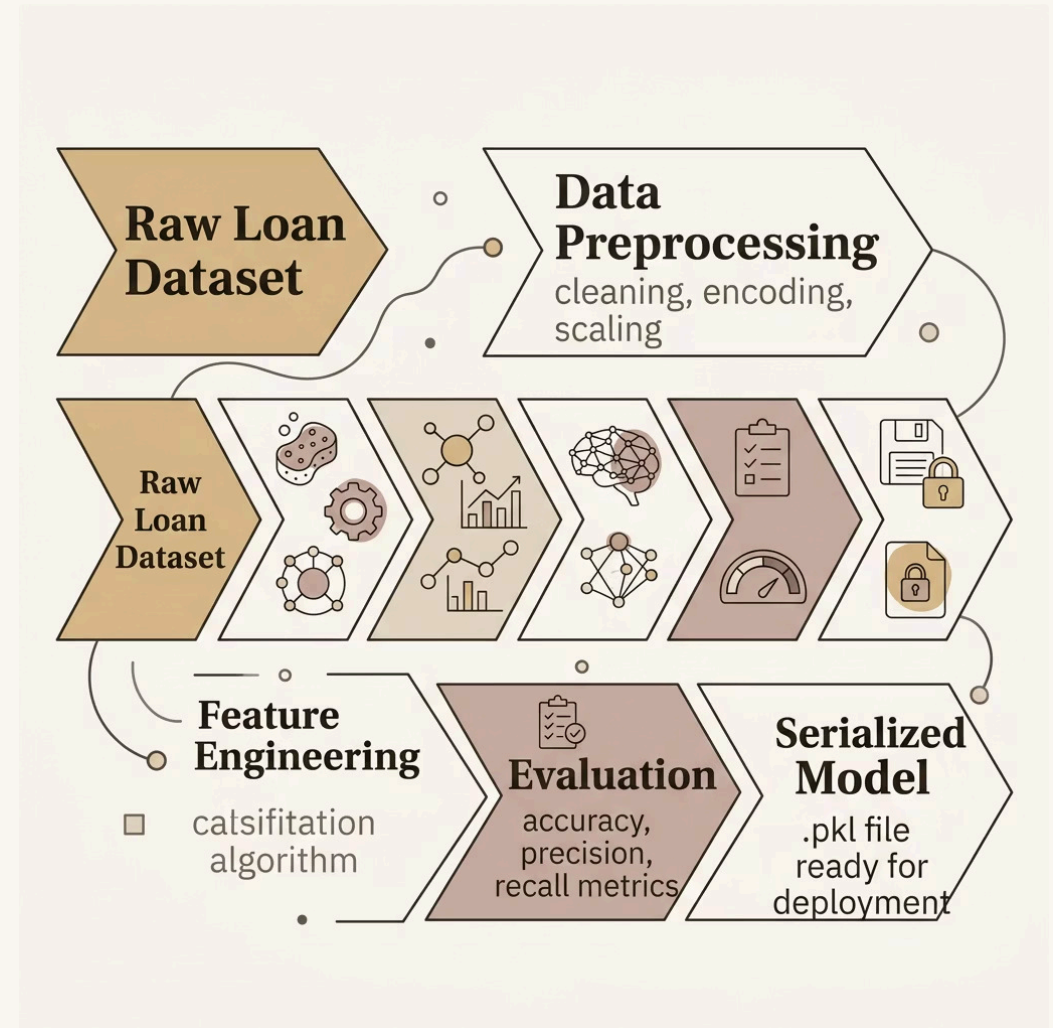
- **Frontend:** Streamlit — rapid, Python-native UI
- **ML Core:** Scikit-learn, Pandas, NumPy
- **Cloud:** AWS EC2 with Security Groups & Nginx
- **Version Control:** Git & GitHub
- **Model Storage:** Pickle (.pkl) serialization

Dataset & ML Model

Training Pipeline

The model is trained on a structured loan dataset covering applicant income, credit history, loan amount, and repayment capacity. Preprocessing handles missing values, categorical encoding, and feature scaling.

- 80/20 train-test split
- Classification algorithm (Logistic Regression / Random Forest)
- Accuracy, Precision & Recall evaluation
- Serialized as .pkl for inference



System Workflow



User Input

Applicant enters loan details — income, amount, credit history.



Streamlit Frontend

Validates and packages data for backend processing.



Python Backend

Processes request, loads serialized ML model.



ML Risk Prediction

Model evaluates eligibility and calculates dynamic interest rate.

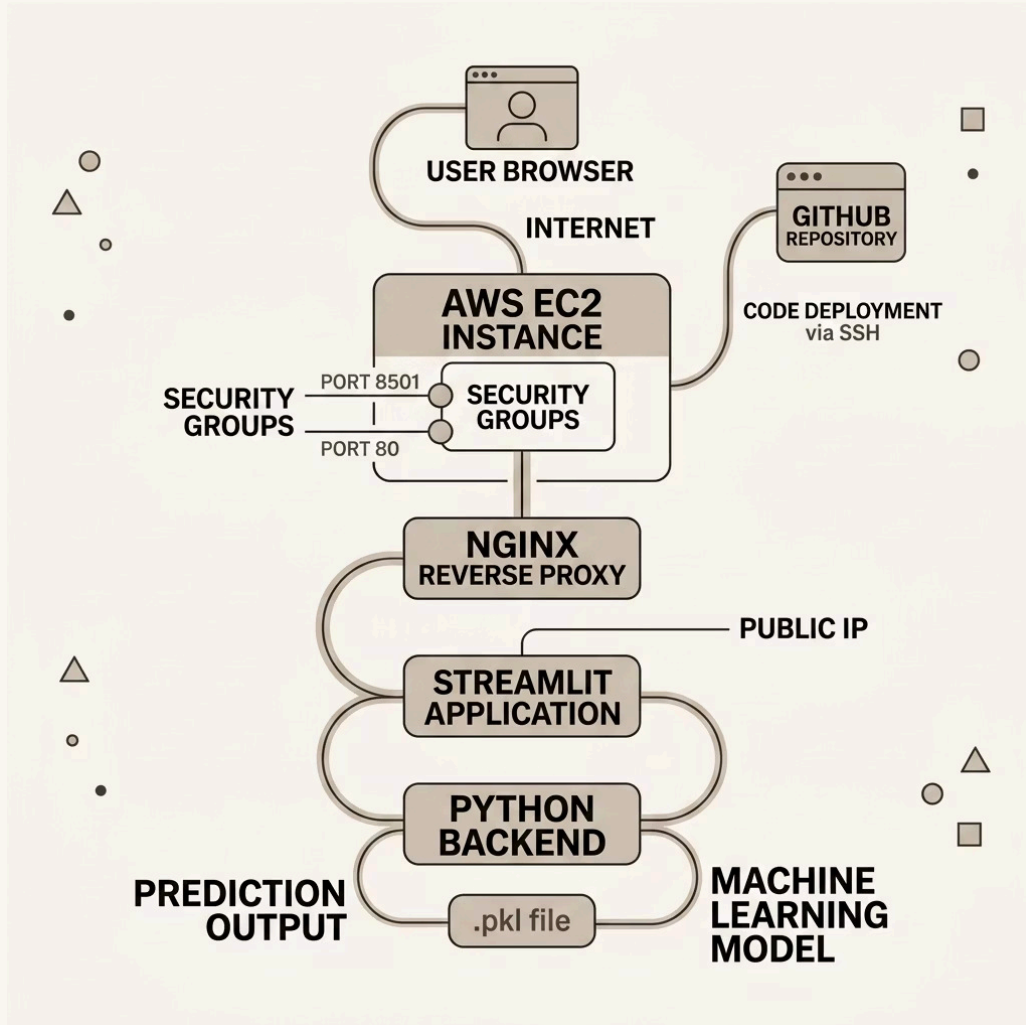


Decision Display

Approval or rejection shown instantly with rate breakdown.

AWS Deployment Architecture

End-to-end cloud infrastructure powering KreditKisan on AWS — from user browser to ML inference engine.



Architecture Highlights

- **AWS EC2:** Primary compute instance hosting the full application stack
- **Security Groups:** Firewall rules allowing only Port 8501 (Streamlit) & Port 80 (Nginx)
- **Nginx:** Optional reverse proxy for production-grade request routing
- **GitHub → EC2:** CI/CD via SSH clone for seamless code deployment
- **Public IP:** Globally accessible endpoint for demo and evaluation
- **.pkl Model:** Serialized ML artifact loaded at runtime for instant inference

AWS Deployment Process

01

Launch EC2 Instance

Ubuntu AMI, t2.micro or higher.

02

Configure Security Groups

Open Port 8501 (Streamlit) and Port 22 (SSH).

03

Connect via SSH

```
ssh -i key.pem ubuntu@public-ip
```

04

Install Python & Dependencies

```
pip3 install streamlit scikit-learn pandas numpy
```

05

Clone GitHub Repository

```
git clone your-repo-url
```

06

Install requirements.txt

```
pip3 install -r requirements.txt
```

07

Train Model / Generate .pkl

Run training script to serialize model.

08

Run Streamlit App

```
streamlit run app.py --server.port 8501
```

09

Expose via Port 8501

Confirm security group allows inbound traffic.

10

Access via Public IP

Open <http://public-ip:8501> in browser.

Advantages, Impact & Future Scope

Why KreditKisan Matters

- **Faster Decisions:** Real-time loan approval replaces days-long manual processes
- **Personalized Pricing:** Interest rates reflect individual risk profiles
- **Cloud Scalability:** AWS infrastructure scales with demand
- **Financial Inclusion:** AI-driven access for underserved borrowers

What's Next

- **Banking APIs:** Live integration with lender systems
- **Blockchain:** Tamper-proof loan verification ledger
- **Mobile App:** Android/iOS version for rural users
- **Deep Learning:** LSTM/Neural networks for richer predictions
- **Multi-language:** Regional language support for India